3.29 Discrete Valuation Rings

A Discrete Valuation Ring (DVR) corresponds to a regular local ring of dimension 1 and therefore regular (or in the case of perfect base field, non-singular) points on a curve. There are a number of equivalent conditions which are summarised here.

Lemma 3.29.1
Let $(A, \mathfrak{m})$ be a local domain such that $\mathfrak{m}$ is finitely generated and $\dim A = 1$. Then for every non-zero ideal $\mathfrak{a}$ there exists some n such that $\mathfrak{m}^n \subseteq \mathfrak{a}$.

Proof. As A is a domain of dimension 1, $\mathfrak{m}$ is its only non-zero prime ideal. Every minimal prime over $\mathfrak{a} \neq 0$ is non-zero, hence equal to $\mathfrak{m}$, so $\sqrt{\mathfrak{a}} = \mathfrak{m}$ by (3.3.46) (the version over minimal primes), i.e. $\mathfrak{a}$ is $\mathfrak{m}$-primary. As $\mathfrak{m}$ is finitely generated there is some n such that $\mathfrak{m}^n \subseteq \mathfrak{a}$ (3.3.48). ∎

Proposition 3.29.2 (DVR Criteria)
Let A be an integral domain which is not a field. Then the following are equivalent.

a) A is a valuation ring whose valuation group is order-isomorphic to $\mathbb{Z}$
b) A is a local principal ideal domain
c) A is a Noetherian local ring with maximal ideal $\mathfrak{m} = (\pi)$ and $\dim A = 1$
d) A is a local ring such that there exists $\pi \in A$ for which every non-zero ideal is of the form $\left(\pi^k\right)$ for some $k \geq 0$ (and these are distinct)
e) There exists a non-unit $\pi \in A$ such that every non-zero element $a \in A$ may be represented as $u\pi^k$ for some $u \in A^\star$ and non-negative integer k.
f) A is a Noetherian valuation ring
g) A is a Noetherian integrally closed local domain with $\dim A = 1$
h) A is a Noetherian regular local ring with $\dim A = 1$

*In this case we say A is a **discrete valuation ring** (or **DVR**) and let $K = \mathrm{Frac}(A)$. For every generator π of $\mathfrak{m}$, the ideals (π^n) are distinct and*

- *for every $x \in K^\star$ there is a unique integer k and unit u such that $x = u\pi^k$, and*
- *there is a well-defined valuation $v : K^\star \to \mathbb{Z}$ determined uniquely by the relation $x = u\pi^{v(x)}$*

Proof. a) $\implies$ b) Consider a valuation $v : K^\star \to \mathbb{Z}$ with $A = \{0\} \cup v^{-1}(\mathbb{Z}_{\geq 0})$. For any non-zero ideal $\mathfrak{a}$ choose non-zero $f \in \mathfrak{a}$ such that $v(f)$ is minimal. For $g \in \mathfrak{a}$ we have $v(g) \geq v(f) \implies v(gf^{-1}) \geq 0 \implies gf^{-1} \in A \implies g \in (f)$. Therefore $\mathfrak{a} = (f)$ is principal. Further by (3.23.2) A is a local ring.

b) $\implies$ c). By (3.20.4) $\dim A = 1$ and by assumption $\mathfrak{m} = (\pi)$. Evidently A is Noetherian.

c) $\implies$ d). The sequence $\left(\pi^k\right)$ is strictly decreasing, for otherwise $\mathfrak{m}^k = \left(\pi^k\right) = \left(\pi^{k+1}\right) = \mathfrak{m}^{k+1}$ for some k and $\dim A = 0$ by (3.27.1).c) $\implies$ a), a contradiction. Assume wlog that $\mathfrak{a}$ is proper and non-zero, so $(\pi^m) = \mathfrak{m}^m \subseteq \mathfrak{a}$ for some m by (3.29.1).

Suppose that $\mathfrak{a} \subseteq \left(\pi^k\right)$ for all k. Then $(\pi^m) \subseteq \mathfrak{a} \subseteq \left(\pi^{m+1}\right) \subseteq (\pi^m)$ which contradicts the sequence being strictly decreasing. Therefore we may choose k maximal subject to $\mathfrak{a} \subseteq \left(\pi^k\right)$; then there exists $a \in \mathfrak{a} \setminus \left(\pi^{k+1}\right)$. Therefore $a = u\pi^k$ and by construction $u \notin (\pi) \implies u \in A^\star \implies \pi^k \in \mathfrak{a}$. Therefore $\mathfrak{a} = \left(\pi^k\right)$ as required.

d) $\implies$ e). By assumption $(a) = \left(\pi^k\right)$ for some integer k, whence by (3.15.3) $a = u\pi^k$ for some $u \in A^\star$. This representation is unique because the ideals $\left(\pi^k\right)$ are all distinct and A is cancellable.

e) $\implies$ a). We first show that k is unique. As A is not a field there is a non-zero non-unit $a = u\pi^k$, with $k \geq 1$, so $\pi \neq 0$. If $u\pi^k = u'\pi^{k'}$ with $u, u' \in A^\star$ and $k < k'$, cancelling π^k gives $u(u')^{-1} = \pi^{k'-k}$, a unit, so π is a unit, a contradiction.

Then for each $a \in A \setminus \{0\}$ let $v(a)$ denote the unique integer such that $a = u\pi^{v(a)}$ for some $u \in A^\star$.

We claim the sequence $\left(\pi^k\right)$ is strictly decreasing, for if not then $\pi^k = w\pi^{k+1}$ for some $w \in A \setminus \{0\}$, cancelling implies π is a unit, a contradiction.

We need to show that v is a valuation. By uniqueness we deduce $v(ab) = v(a) + v(b)$. Suppose that $a, b, a+b \in A \setminus \{0\}$. Then $\left(\pi^{v(a+b)}\right) = (a+b) \subseteq \left(\pi^{\min(v(a), v(b))}\right)$. As the sequence $\left(\pi^k\right)$ is strictly decreasing we find $v(a+b) \geq \min(v(a), v(b))$ as required. We extend v to $K^\star$ by $v(ab^{-1}) := v(a) - v(b)$ for $a, b \in A \setminus \{0\}$. This is well defined: if $ab^{-1} = a'b'^{-1}$ then $ab' = a'b$, so $v(a) + v(b') = v(a') + v(b)$ by multiplicativity. It is a homomorphism $K^\star \to \mathbb{Z}$, and it is ultrametric: for $x, y \in K^\star$ with $x + y \neq 0$ write $x = ad^{-1}$, $y = bd^{-1}$ over a common denominator $d \in A \setminus \{0\}$, so $v(x+y) = v(a+b) - v(d) \geq \min(v(a), v(b)) - v(d) = \min(v(x), v(y))$. It is onto $\mathbb{Z}$ as $v(\pi) = 1$. Then $v(ab^{-1}) \geq$

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$0 \iff v(a) \geq v(b) \iff (a) \subseteq (b) \iff b \mid a \iff ab^{-1} \in A$. Further $v(a) = 0 \iff a \notin (\pi) \iff a \in A^\star$. Therefore A is the valuation ring associated to v.

b) $\iff$ f). This is the last statement of (3.23.21): a Noetherian valuation ring is the same as a local principal ideal domain.

f) $\implies$ g). A valuation ring is integrally closed (3.23.2) and we've already shown A is local with dimension 1 (via f) $\implies$ b) $\implies$ c)).

g) $\implies$ c). Let $0 \neq a \in \mathfrak{m}$. As $\dim A = 1$ from (3.29.1) we know $\mathfrak{m}^n \subseteq (a)$ for some $n \geq 1$. Choose n minimal and $b \in \mathfrak{m}^{n-1} \setminus (a)$.

Let $x = a^{-1}b \in K = \mathrm{Frac}(A)$. Evidently $x \notin A$, for otherwise $b \in (a)$; further $x\mathfrak{m} = a^{-1}b\mathfrak{m} \subseteq a^{-1}\mathfrak{m}^n \subseteq a^{-1}(a) = A$ is an ideal of A.

We claim that $x\mathfrak{m} = A$. Suppose not then $x\mathfrak{m} \subseteq \mathfrak{m}$ by (3.3.37). Then $\mathfrak{m}$ is a faithful finite $A[x]$-module and x is integral over A (3.22.3). By assumption A is integrally closed so $x \in A$ which is a contradiction. Therefore $x\mathfrak{m} = A \implies \mathfrak{m} = x^{-1}A$. As $1 \in A$ we get $x^{-1} \in x^{-1}A = \mathfrak{m} \subseteq A$, so $x^{-1} \in A$ and $\mathfrak{m} = (x^{-1})$ is a principal ideal.

c) $\implies$ h). By assumption A is a Noetherian local ring with $\dim A = 1$. So it remains to show that $\dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2 = 1$. As $\mathfrak{m}$ is principal then $\dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2 \leq 1$. If it is zero then $\mathfrak{m} = \mathfrak{m}^2$ and $\dim A = 0$ by (3.27.1), a contradiction.

h) $\implies$ c). By assumption $\dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2 = 1$ and so by (3.19.7) $\mathfrak{m} = (\pi)$ for some $\pi \in \mathfrak{m} \setminus \mathfrak{m}^2$.

The final statements have already been proven. ∎

Lemma 3.29.3 (Valuation and Powers of the Maximal Ideal)
Let $(A, \mathfrak{m})$ be a DVR with valuation v, and $k \geq 0$ an integer. For $0 \neq a \in A$:

a) $v(a) \geq k \iff a \in \mathfrak{m}^k$;
b) $v(a) = k \iff a \in \mathfrak{m}^k \setminus \mathfrak{m}^{k+1}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Fix a uniformiser π (so $\mathfrak{m} = (\pi)$, hence $\mathfrak{m}^k = (\pi^k)$ (3.29.2)), and write $a = u\pi^{v(a)}$ for a unit $u \in A^\star$ (3.29.2).

a). $a \in \mathfrak{m}^k = (\pi^k) \iff \pi^k \mid a = u\pi^{v(a)} \iff \pi^k \mid \pi^{v(a)}$ (as u is a unit) $\iff v(a) \geq k$ (as the ideals (π^n) are strictly decreasing (3.29.2)).

b). By a), $a \in \mathfrak{m}^k \setminus \mathfrak{m}^{k+1} \iff v(a) \geq k$ and not $(v(a) \geq k+1) \iff v(a) = k$. ∎

Corollary 3.29.4
Let A be a principal ideal domain which is not a field. Then every localisation of A at a non-zero prime ideal is a DVR.

In particular $\mathbb{Z}_{(p)}$ and $k[X]_{(f)}$ are DVRs where p is prime and $f(X)$ is an irreducible polynomial.

Lemma 3.29.5 (A DVR is Maximal Among Valuation Subrings of its Fraction Field)
Let A be a DVR with fraction field K and B a valuation ring of K with $A \subseteq B$. Then $A = B$ or $B = K$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Suppose $B \neq K$. A valuation ring of K which is a field equals K (if $x \in K \setminus B$ then $x^{-1} \in B$ is a non-unit of B, as otherwise $x \in B$), so B is not a field and $\dim B \geq 1$. As $\dim A = 1$ (3.29.2) and $A \subseteq B$, (3.23.14) gives $\dim B \leq \dim A = 1$ with equality iff $A = B$. Hence $\dim B = \dim A$ and $A = B$. ∎

Lemma 3.29.6 (Directed Union of Valuation Rings with a Common Uniformiser)
Let $(K_i)_{i \in I}$ be a directed system of fields with union $K := \bigcup_i K_i$, and for each i let $R_i \subset K_i$ be a valuation ring with $R_i \subseteq R_l$ whenever $K_i \subseteq K_l$. Suppose in addition every $\mathfrak{m}_{R_i}$ is principal, generated by a single common element $0 \neq \pi \in \bigcap_i R_i$. Then $R := \bigcup_i R_i$ is a valuation ring of K, with $\mathfrak{m}_R = \pi R \neq 0$; in particular $R \subsetneq K$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

For $K_i \subseteq K_l$ (so $R_i \subseteq R_l$), $\mathfrak{m}_{R_i} = \pi R_i \subseteq \pi R_l = \mathfrak{m}_{R_l}$, giving R_l dominates R_i (3.23.4) so (3.23.7) applies, and $\mathfrak{m}_R = \bigcup_i \mathfrak{m}_{R_i} = \bigcup_i \pi R_i = \pi R$.

As $\pi \in \mathfrak{m}_{R_i}$ is a non-unit of every R_i, $\pi^{-1} \notin R_i$ for every i (well-defined, as $0 \neq \pi \in K_i \subset K$), hence $\pi^{-1} \notin R = \bigcup_i R_i$; as $\pi^{-1} \in K$, $R \subsetneq K$. ∎

Proposition 3.29.7
Let A be a discrete valuation ring with unique maximal ideal $\mathfrak{m}$. Then there exists a canonical group isomorphism

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$$(\mathfrak{m})^- : \mathbb{Z} \xrightarrow{\sim} \mathcal{I}_A$$

which is the unique group homomorphism mapping 1 *to* $\mathfrak{m}$ *(and* 0 *to* A*. In particular every fractional ideal is invertible.*

Write the inverse group isomorphism by $v_\mathfrak{m}$*, which satisfies*

a) $\mathfrak{m}^{v_\mathfrak{m}(\mathfrak{a})} = \mathfrak{a}$
b) $v_\mathfrak{m}(\mathfrak{m}^k) = k$

A fractional ideal I *is contained in* A *if and only if* $v_\mathfrak{m}(I) \geq 0$*.*

Proof. By (3.29.2) we may suppose $\mathfrak{m} = (\pi)$. Evidently (π^k) is a fractional ideal for every integer k and $(\pi^k)(\pi^l) = (\pi^{k+l})$, which gives the required homomorphism $k \to (\pi^k)$.

Suppose the kernel is non-trivial then there exists k which maps to A, and we may assume that k is positive (by considering the inverse). This means precisely that $\mathfrak{m}^k = A$, which may only happen when $k = 0$ because $\mathfrak{m}$ is proper.

Let I be an arbitrary fractional ideal then $aI \subseteq A$ for some $a \in A$. By (3.29.2) $a = u\pi^l$ and $aI = (\pi^k)$ for some positive integers l and k. This means $(\pi^l)I = (\pi^k)$ and therefore $I = (\pi^k)(\pi^{-l}) = (\pi^{k-l})$. This shows the homomorphism is surjective and therefore an isomorphism.

Evidently $(\pi^k) \subset A \iff k \geq 0$ since $\pi \in A$ and $\pi^{-1} \notin A \implies \pi^{-k} \notin A$. ∎

Proposition 3.29.8
Let A *be a discrete valuation ring and* I, J *fractional ideals. Then the following are equivalent*

a) $I \subset J$
b) J *divides* I *(that is there exists an ideal* $\mathfrak{a}$ *such that* $\mathfrak{a}J = I$*)*
c) $v_\mathfrak{m}(I) \geq v_\mathfrak{m}(J)$

Proof. Using the results of (3.29.7) we may show (eliding the subscript $\mathfrak{m}$)

a) $\iff$ c) $I \subset J \iff J^{-1}I \subset A \iff v(J^{-1}I) \geq 0 \iff v(I) \geq v_\mathfrak{m}(J)$

b) $\implies$ c) $v(I) = v(\mathfrak{a}) + v(J) \geq v(J)$

c) $\implies$ b) Define $\mathfrak{a} = \mathfrak{m}^{v(I)-v(J)}$ then evidentally $v(\mathfrak{a}J) = v(I)$ whence $I = \mathfrak{a}J$. ∎

Lemma 3.29.9 (Intersection of the Powers of the Maximal Ideal)
Let A *be a DVR with maximal ideal* $\mathfrak{m}$*. Then* $\bigcap_{n\geq 0} \mathfrak{m}^n = 0$*.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Let $0 \neq a \in A$. By (3.29.3) $a \in \mathfrak{m}^n \iff v(a) \geq n$, so $a \notin \mathfrak{m}^{v(a)+1}$ and in particular $a \notin \bigcap_n \mathfrak{m}^n$. ∎

Lemma 3.29.10 (Ramification and Injectivity/Surjectivity of Local Homomorphisms)
Let $(A, \mathfrak{m})$*,* $(B, \mathfrak{n})$ *be DVRs with valuations* v_A, v_B *(3.29.2), and* $\phi : A \to B$ *a local ring homomorphism (that is* $\phi(\mathfrak{m}) \subseteq \mathfrak{n}$*). Fix a uniformiser* π *of* A *and set* $e := v_B(\phi(\pi)) \in \mathbb{N} \cup \{\infty\}$ *(the* ***ramification index*** *of* ϕ*; independent of the choice of* π*, with* $e = \infty$ *iff* $\phi(\pi) = 0$*). Then:*

a) ϕ *is injective if and only if* $e < \infty$*;*
b) if ϕ *is surjective, then* $e = 1$*; in particular, by a),* ϕ *is then also injective – so* ϕ *is surjective if and only if it is bijective.*

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Independence of π. If $\pi' = u\pi$ for a unit $u \in A^\star$, then $\phi(\pi') = \phi(u)\phi(\pi)$ with $\phi(u) \in B^\star$ (units map to units under any ring homomorphism, and $\phi(u) \notin \mathfrak{n}$, as otherwise $1 = \phi(u^{-1})\phi(u) \in \mathfrak{n}$, absurd), so $v_B(\phi(\pi')) = v_B(\phi(\pi))$.

a). $\ker\phi$ is a prime ideal of A (the kernel of a ring homomorphism into the domain B), so $\ker\phi \in \{0, \mathfrak{m}\}$ (3.29.2). If $e = \infty$: $\phi(\pi) = 0$, so $\pi \in \ker\phi$, and, as $\mathfrak{m} = (\pi)$, $\mathfrak{m} \subseteq \ker\phi$; as $\ker\phi \neq A$ ($\phi(1) = 1 \neq 0$), $\ker\phi = \mathfrak{m} \neq 0$: ϕ is not injective. If $e < \infty$: $\phi(\pi) \neq 0$, so $\pi \notin \ker\phi$, giving $\ker\phi \neq \mathfrak{m}$, so $\ker\phi = 0$: ϕ is injective.

b). Suppose ϕ is surjective. Let ϖ be a uniformiser of B; by surjectivity, $\varpi = \phi(a)$ for some $a \in A$, necessarily $a \neq 0$ (as $\varpi \neq 0$). Write $a = u\pi^k$ for a unit $u \in A^\star$ and $k := v_A(a) \geq 0$ (3.29.2). If $e = \infty$: $\phi(a) = \phi(u)\phi(\pi)^k$ is $\phi(u)$ (a unit, as units map to units) if $k = 0$, or 0 if $k \geq 1$; neither equals ϖ ($v_B(\varpi) = 1 \neq 0$, $\varpi \neq 0$), a contradiction. So $e < \infty$, and

$$1 = v_B(\varpi) = v_B(\phi(u)) + k \cdot v_B(\phi(\pi)) = 0 + ke = ke,$$

forcing $e = 1$. ∎

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